AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2017

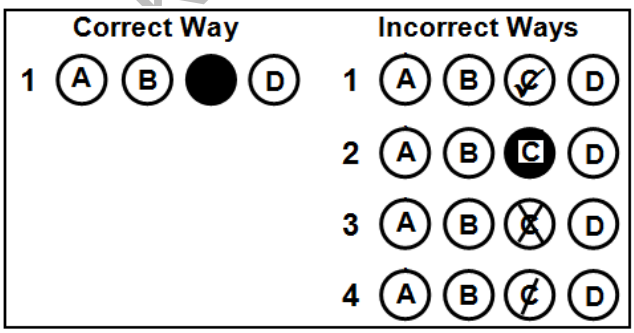
Physics Paper I

Time: 35 minutes Marks: 25

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. The total number of basic physical quantities are

   A. 3
   B. 5
   C. 7
   D. 9

2. The density of pure water is 1000 kg/m^3. This value in scientific notation can be written as

   A. 1.0×10^{-3} kg/m^3
   B. 10.0×10^{-3} kg/m^3
   C. 1.0×10^{3} kg/m^3
   D. 10.0×10^{3} kg/m^3

3. Which of the following measuring instruments is used to find the diameter of a metallic wire?



A B C D

4. If a body is falling freely, its motion will be

   A. linear.
   B. random.
   C. periodic.
   D. vibratory.

5. Which of the following statements is CORRECT about force and velocity?

   A. Both are scalar quantities.
   B. Both are vector quantities.
   C. Force is a vector quantity and velocity is a scalar quantity.
   D. Force is a scalar quantity and velocity is a vector quantity.

6. The gradient of a line on the distance-time graph represents

   A. time.
   B. speed.
   C. acceleration.
   D. displacement.

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7. If a body is taken to the surface of the moon, then which of the following physical quantities of the body will change?

   A. Mass
   B. Length
   C. Weight
   D. Density

8. If the force acting on a body is doubled, then the acceleration produced in it will be

   A. halved.
   B. tripled.
   C. doubled.
   D. quadrupled.

9. Two equal but opposite parallel forces acting at different points on a body would form a

   A. torque.
   B. couple.
   C. moment arm.
   D. line of action.

10. The product of force and moment arm of the force is called

    A. torque.
    B. couple.
    C. equilibrium.
    D. centre of gravity.

11. A point where the whole weight of a body acts vertically downward is called the

    A. centre of Earth.
    B. point of contact.
    C. centre of gravity.
    D. point of applied force.

12. The value of acceleration due to gravity ‘g’ decreases with the

    A. decrease in mass.
    B. increase in weight.
    C. decrease in density.
    D. increase in altitude.

13. The moon revolves around the Earth due to the gravitational force exerted on the moon by the

    A. Sun.
    B. Earth.
    C. space.
    D. satellite.

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14. If an astronaut attain a height equal to the radius of the Earth, then his weight related to the Earth surface will become

    A. half.
    B. twice.
    C. one-third.
    D. one-fourth.

15. If the kinetic energy of a 200 kg object is 10,000 J, then the velocity of the object will be

    A. 0.01 m/s.
    B. 0.02 m/s.
    C. 10 m/s.
    D. 100 m/s.

16. The vertical height of a staircase is 5 m. If a student of mass 51 kg runs up the stairs in 10 s, then the power exerted by the student will be

    (**Note:** Take acceleration due to gravity 'g' = 10 m/s^2.)

    A. 2.55 W
    B. 25.5 W
    C. 255 W
    D. 2550 W

17. If 25 N force is applied to drag a table in the horizontal direction through a distance of 5 m, then the magnitude of work done on the table will be

    A. 0.2 J
    B. 5 J
    C. 20 J
    D. 125 J

18. If a person exerts force which causes a car to move forward, then this is an example of

    A. power.
    B. efficiency.
    C. work done.
    D. rotatory motion.

19. According to the kinetic molecular model of matter, molecules of a substance

    A. are in the state of rest.
    B. are in the state of motion.
    C. have constant momentum.
    D. have same velocity during collision.

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20. The property of matter by which it restores its length, shape or volume after the removal of the deforming force is called

A. stress.
B. strain.
C. elasticity.
D. elastic limit.

21. The quantity that determines the average kinetic energy of the molecules of a substance is known as

A. heat.
B. temperature.
C. latent heat of fusion.
D. latent heat of vaporisation.

22. The escape of high kinetic energy molecules in the form of vapours from the surface of a liquid without heating is known as

A. fusion.
B. boiling.
C. evaporation.
D. condensation.

23. The table given below shows four different substances of equal masses with their values of specific heat capacity. If the same amount of heat is given to all, then which substance will show the maximum rise in temperature?

	Substance of Equal Mass	**Specific Heat Capacity (J/kg.K)**
A	Brick	900
B	Carbon	121
C	Alcohol	2500
D	Aluminium	903

24. All of the following are factors on which rate of evaporation depends EXCEPT on the

A. surface area.
B. surface temperature.
C. colour of the surface.
D. pressure at the surface.

25. All of the following show transfer of heat by convection EXCEPT the use of

A. fans to dry off sweat.
B. hot air to fly off gliders.
C. gas heaters to heat up rooms.
D. heating pads to relax muscles.

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